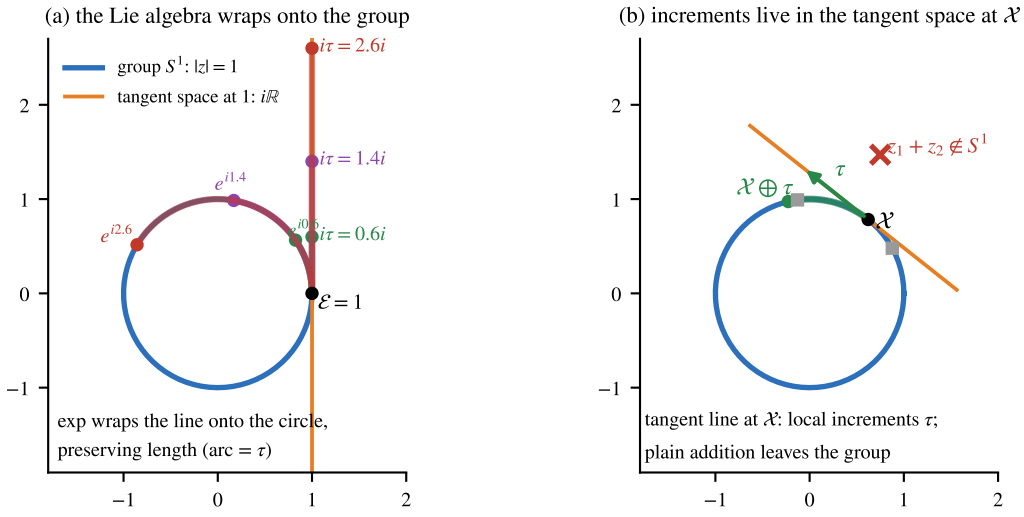
(a) the Lie algebra wraps onto the group
group S^1: $|z| = 1$
tangent space at 1: $i\mathbb{R}$
$i\tau = 2.6i$
$i\tau = 1.4i$
$i\tau = 0.6i$
$e^{i1.4}$
$e^{i2.6}$
$e^{i0.6}$
$\mathcal{E} = 1$
exp wraps the line onto the circle,
preserving length (arc = τ)
(b) increments live in the tangent space at $\mathcal{X}$
$z_1 + z_2 \notin S^1$
τ
$\mathcal{X} \oplus \tau$
$\mathcal{X}$
tangent line at $\mathcal{X}$: local increments τ;
plain addition leaves the group